

Topic: Memory Segmentation in 8086

Q. Explain the concept of Memory Segmentation in the 8086 microprocessor. How is a 20-bit physical address calculated using a 16-bit segment register and a 16-bit offset

[DEF] Definition to Remember

Memory Segmentation divides the 1 MB of physical memory of the 8086 into logical 64 KB blocks (segments). Because internal registers are only 16-bit (max 64 KB), the CPU mathematically combines a 16-bit Segment Base and a 16-bit Offset to generate the required 20-bit physical address.

Memory Segmentation in 8086

The CPU manages these memory blocks using four specialized 16-bit Segment Registers:

2. Logical vs Physical Address

* Logical Address: Format used by programmers (Segment : Offset). e.g., 1000H:0002H.

3. Calculating the 20-bit Physical Address

The Bus Interface Unit (BIU) performs this calculation:

Calculation Example:

1. Shift Segment by 10H: $2000H \times 10H = 20000H$ (20-bit Base)

4. Advantages of Segmentation

* Code Relocation: Programs can be moved anywhere in memory by changing only the Segment Register; internal offset addresses remain unchanged.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. 8086 has a 20-bit address bus (1 MB memory) but only 16-bit internal registers.
2. Segmentation divides memory into 64 KB logical blocks to solve this addressing mismatch.
3. Four Segment Registers: CS (Code), DS (Data), SS (Stack), ES (Extra).
4. Physical Address is calculated by the BIU combining a Segment Base and an Offset.
5. Formula: $\text{Physical Addr} = (\text{Segment Value} \times 10H) + \text{Offset}$.
6. Provide the calculation example (e.g., CS=2000H, IP=1234H $\rightarrow$ Base=20000H $\rightarrow$ Result=21234H).
7. Advantages: Easy program relocation and separation of code/data/stack.

[RECALL] Quick Recall

Segmentation -> 1 MB Memory / 16-bit registers -> Divides into 64KB blocks (CS, DS, SS, ES) -> Physical Addr = (Segment x 10H) + Offset -> Pros: Relocation, Code/Data separation